# MODUPE AWOJUYIGBE

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## SUMMARY

Computer Engineering student experienced in building practical, scalable tools with Python, C++, cloud technologies, and AI. I work across software, automation, and system optimization to solve complex problems, improve user experience, and create real impact.

## EDUCATION

**The University of Texas at San Antonio** *Expected: May 2027*

- Bachelor of Science in **Computer Engineering/** Minor in **Artificial Intelligence** – GPA: **3.7/4.0**

**Achievements:** President's List, Dean's List, Women in Toys Scholar, Klesse College Scholar, NSBE Scholar, USAA Scholar, Society of Women Engineers (SWE), herScript, Rewriting the Code

**Coursework:** Digital System Design, C++ and Data Structures, Engineering Analysis II, Micro Computer Systems

## SKILLS

- **Languages & Frameworks:** Python (Pandas, Selenium), C++, C, JavaScript (Node.js, Express.js),PyTorch, SQL, MATLAB, HTML/CSS.
- **Cloud & AI**: Google Cloud (Vertex AI, Gemini), Generative AI, LLMs, Firebase, AI Agent Development, MCP (model context protocol), Vector Databases (TurboPuffer), Automation Scripting.
- **Systems, Hardware & Tools:** SCADA Systems, Embedded Systems (PIC), DNP3, Agile Methodology, Git, Circuit Analysis, Root Cause Analysis.

## EXPERIENCE

**EDP Renewables**

*Engineering Intern, Houston, TX* **June 2025 – August 2025**

- Designed a standardized data template that **reduced project setup delays by 70%**, streamlining **cross-functional** coordination between contractors and engineering teams.
- Engineered a Python automation tool (Selenium) to aggregate weather data, reducing reporting time by **50%**.
- Reduced company spend by **$200,000** in operational costs via automated tools and data logging solutions that improved workflow and visibility.

**The University of Texas at San Antonio**

*Teaching Assistant / Peer Coach, San Antonio, TX* **August 2024 – Present**

- Mentored **50+** students in Calculus and programming, translating complex engineering concepts into real-world applications, which improved student pass rates by **60%.**

**Scholist App**

*Full Stack Developer Intern, Dallas, TX* **May 2023 – August 2023**

- Architected a **Full Stack** scholarship platform within an **Agile** environment, transforming **Figma** prototypes into responsive **JavaScript** interfaces and scalable **Node.js** backends.
- Engineered a high-performance **RAG** recommendation system using **TurboPuffer** for vector storage and **Google Vertex AI**, achieving sub-millisecond retrieval speeds that reduced user churn by **50%** and increased paid conversions by **80%**

## PROJECTS

**Robotic Arm Project** **Spring 2026**

- Programmed an Arduino-based robotic arm in **C++** with ultrasonic sensors, implementing PID logic to detect objects and improve pickup accuracy by **60%**

**Real-Time Voice Captioning and Translation System** **Spring 2025**

- Built a real-time captioning and translation system using a **PIC microcontroller** and **Python** to convert speech into formatted text on an LCD.
- Developed a Python speech-to-text pipeline using **Cloud Translation API** and **Gemini** for contextual understanding and multilingual translation across 100+ languages.

**Autonomous Line Tracking Vehicle** **Fall 2024**

- Engineered a fully autonomous mini vehicle utilizing infrared sensors and **C++** programming to enable precise path detection and following capabilities.
- Designed and implemented a control system architecture integrating microcontrollers with motor drivers to achieve responsive navigation with **<2ms** latency.

## LEADERSHIP

**National Society of Black Engineers (NSBE) –** *Senator* **May 2025 - Present**

- Represent the UT San Antonio chapter at regional and national meetings, advocate for student needs and initiatives
- Planned and executed outreach, programming, and professional development events, boosting chapter participation by **30%**